

LESSON 1.1

USING MULTIPLE REPRESENTATIONS TO INVESTIGATE PATTERNS



LESSON INTRODUCTION

MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.

MA.5.AR.2 Given a rule for a numerical pattern, use a two-column table to record the inputs and outputs.

MA.5.GR.4.2 Represent mathematical and real-world problems by plotting points in the first quadrant of the coordinate plane and interpret 3 coordinate values of points in the context of the situation.

LEARNING TARGETS

- I can use multiple representations to investigate patterns found in tables or lists.
- I can plot points on the coordinate plane.

BENCHMARK COHERENCE

WORKING TOWARD

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

ESSENTIAL QUESTIONS

- How do you plot points on the coordinate plane?
- How can you write a rule to describe a pattern as an expression?
- How do equations, tables, and graphs help you see patterns and interpret data?

LESSON NARRATIVE

This lesson activates students' prior knowledge of patterns and representing relationships on a coordinate plane. Through guided instruction and practice, students investigate how to complete a table or a list to extend a pattern, then create an expression that represents the pattern. Students then use their knowledge of patterns to find related terms and use the terms as ordered pairs on a coordinate plane.

COMPONENT SUMMARY

1.1.1 WARM-UP (~5 MIN.)

Students analyze and extend a visual pattern

1.1.3 GUIDED PRACTICE (15-20 MIN.)

Students practice identifying a cost pattern, writing an equation, and graphing on the coordinate plane

1.1.2 GUIDED INSTRUCTION (15-20 MIN.)

Identify and write an equation to represent the relationship between oranges and cartons, then graph the pattern on the coordinate plane

1.1.4 WRAP-UP (~5 MIN.)

Self-assessment on identifying, writing, and extending patterns

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Term, Pattern
- Equation
- x-axis, y-axis
- Ordered pair
- Origin

MATERIALS

- (Optional) Chart paper
- (Optional) Carton of orange juice
- (Optional) Math manipulatives or drawing materials for creating/ extending patterns

ADDITIONAL PREPARATION

- (Optional) On chart paper, prepare a large coordinate plane with labels for the origin, x-axis, and y-axis. This anchor chart can be referenced throughout the lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may attempt to use addition instead of multiplication to find the patterns in the table.
- Students may struggle with connecting the coordinate pairs from the table to the coordinate plane points.
- Students may struggle with plotting points on the coordinate plane, confusing the x- and y-values in tables and on a graph.
- Students may not identify a rule that shows how the number pairs are related.

THINGS TO CONSIDER

- To help students identify a rule that shows how the number pairs are related, consider having students test the rule that they identified on several number pairs to determine if it works for all pairs or only the one that they selected.
- Use and encourage the correct use of vocabulary related to coordinate planes.
- Be flexible with the use of a calculator. Consider student learning needs if a calculator could remove arithmetic as the barrier.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 1.1.2 Guided Instruction, students engage in Notice and Wonder literacy routines as they look for and identify patterns, then express them as equations and graph their relationship on a coordinate plane. Encourage students to share their ideas both verbally and in written form, then add to or revise their thinking based on others' responses.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

Throughout all sections of this lesson, students are actively engaged in seeking patterns, extending patterns, and looking for relationships in real-world situations. Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts as exemplified in 1.1.2 Guided Instruction.

DIFFERENTIATE INSTRUCTION

For 1.1.1 Warm-Up, consider having students draw, write, or annotate on the visual pattern as they reason how many triangles will be in the given rows, or use math manipulatives to express the growth from term to term. Students can also draw visual models to better see the relationship between oranges and cartons in 1.1.2 Guided Instruction.

The Wrap-Up is an opportunity for students to reflect on learning from the lesson activities. Consider encouraging students to go back through the student Workbook and place checks (understanding), question marks (need more clarification), and smaller statements next to specific activities to foster metacognitive awareness of learning.

1.1.1 WARM-UP: VISUAL PATTERNS

Component Overview: Display the visual pattern for all students to see, optionally including access to math manipulative and/or drawing materials to create and extend the pattern. Have students individually answer the questions. If time, elicit student responses and reasoning as a class to clarify the thought process about how the terms in the pattern are related.

TEACHER MOVES

Notice and Wonder

Provide students individual think time to create, notice, and wonder about the visual pattern. Prompt students to think about the relationship between terms by asking students to consider how the pattern changes from one term to the next using “term.” As students explain their thinking, consider highlighting responses using the word “row” to help them clarify their reasoning on how the pattern grows by row.

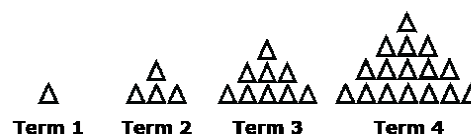
DIFFERENTIATED INSTRUCTION

Encourage students to draw, write, or annotate on the visual pattern as they reason how many triangles will be in the given rows. As students share their responses for questions 2-4, consider writing their ideas on a shared classroom board for everyone to see.



1.1.1 Warm-Up: Visual Patterns

The diagram shows four terms of a visual pattern.



1. What do you notice about the pattern?

Answers will vary. Starting at term 1, the total number of triangles was found by adding 3, 5, and 7. The total number of triangles can be found by multiplying the term number by the number of rows.

2. How many rows of triangles do you think will be in Term 9?
9
3. How many of the small triangles do you think will be in the pattern for Term 9?
81
4. How many of the small triangles do you think will be in the pattern for Term 15?
225



1.1.2 Guided Instruction: Using Multiple Representations to Investigate Patterns

1. A team of mathematicians at an orange juice manufacturer in Florida is collecting information on how many oranges are needed to make orange juice.

The table shows the information the mathematicians have recorded so far.

Cartons of Orange Juice	Average Number of Oranges Needed
2	40
3	60
5	100

- a. What do you notice about the relationship between the cartons of orange juice and the number of oranges needed?
Answers will vary. It takes 20 oranges to make a carton of orange juice.
- b. What is the average number of oranges needed for 9 cartons of orange juice?
180
- c. A healthy, mature orange tree produces between 200 and 360 oranges. What is the range of the number of cartons of orange juice that can be made using oranges from a healthy, mature orange tree?
The range is 10 – 18 cartons.
- d. Write an equation that represents the relationship between the number of cartons of orange juice, x , and y , the average number of oranges needed.
 $y = 20x$

1.1.2 GUIDED INSTRUCTION: USING MULTIPLE REPRESENTATIONS TO INVESTIGATE PATTERNS

Component Overview: In this Guided Instruction, students will learn to use the patterns found in table values from a real-world context to write an equation and create a graphical representation of the context that can be used to interpret the meaning of ordered pairs.

TEACHER MOVES

Consider introducing students to the context by showing them a carton of orange juice and asking them if they ever wondered how many oranges it takes to make a carton of orange juice. Then, use the Poll the Class routine for student responses.

Consider using Notice and Wonder routines for questions 1a and 1b. Elicit responses that notice the relationship between cartons of orange juice and oranges needed and how many oranges are needed per carton. Ask questions like:

- What do you notice about the number of orange juice cartons on the table?
- How does the average number of oranges needed change as the cartons of orange juice change?
- How can we describe how the pattern changes with more cartons?

1.1.2 GUIDED INSTRUCTION: USING MULTIPLE REPRESENTATIONS TO INVESTIGATE PATTERNS (CONTINUED)

ENRICHMENT

Position students to use the table, responses from 1a, and reasoning from 1b as they write the equations in question 1d. Consider asking questions to scaffold learning, including:

- How do we know how many oranges we need per carton? How is that helpful in writing the equation to represent the relationship?
- Why do we use multiplication instead of addition in our equation?

QUESTIONING

As an extension and to further understanding of using the equation to represent relationships, ask students to use the equation in 1d to determine how many cartons 100, 150, or any other value of oranges would create.

DIFFERENTIATED INSTRUCTION

As the class engages with the definitions and descriptions in question 1e, consider using an anchor chart or other visual with labels for each part of the coordinate plane. This learning approach can serve as a review of needed vocabulary, including origin, x-axis, y-axis, and ordered pair.

- e. Use your equation or the pattern to complete the table.

Cartons of Orange Juice	Average Number of Oranges Needed
1	20
2	40
3	60
4	80
5	100

Each pair of values in the table can be plotted on the coordinate plane. The coordinate plane has two axes. The **x-axis** is the horizontal axis and the **y-axis** is the vertical axis.

Each pair of values becomes an **ordered pair**, where the **x-value** is the first number in the ordered pair, and the **y-value** is the second number in the ordered pair.

For example, the mathematicians found that making 2 cartons of orange juice usually uses 40 oranges.

- f. Complete the statement.

The ordered pair that represents this relationship is

- (2, 40).
- (40, 2).

To plot the **x-value**, begin at the **origin**, (0,0), then move

- up 2 units.
- right 2 units.
- right 40 units.
- up 40 units.

To plot the **y-value**,

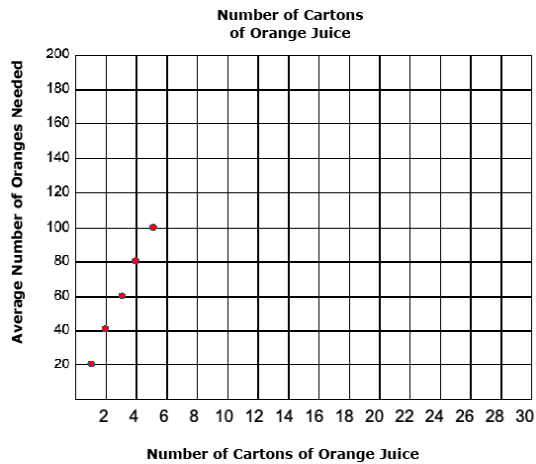
move

- up 2 units
- right 2 units
- right 40 units
- up 40 units

from the **x-value**.

- g. After discussing with your partner, write a description of how to plot the point for 1 carton of orange juice.
Answers will vary. Start at the origin, move right 1, then up 20.

- h. Plot each point from the table on the coordinate plane below. *Answer shown on graph.*



- i. Use the table, equation, or graph to complete the ordered pair.
 (9, 180)
- j. What does each number in the ordered pair represent?
*9 represents the number of cartons of orange juice.
 180 represents the number of oranges needed.*

1.1.2 GUIDED INSTRUCTION: USING MULTIPLE REPRESENTATIONS TO INVESTIGATE PATTERNS (CONTINUED)

ENRICHMENT

Question 1g relies on students identifying that one carton of orange juice requires 20 oranges to write the coordinate pair (1,20). Prompt students to identify this by asking:

- Earlier in the lesson, how did you use the table to determine how many oranges are needed for every carton?
- How can we describe the movement to plot this point on the coordinate plane, starting at the origin?

TEACHER MOVES

For question 1h, consider modelling how to graph the point or have students volunteer to graph the point. As a kinesthetic and auditory entry point, students can read the statement as they start at the origin, then physically move a marker across the x- and then y-axis on a coordinate plane paper and/or the class anchor chart. This strategy can also help highlight the importance of moving horizontally on the x-axis first, then vertically on the y-axis second.

ENRICHMENT

For questions 1h-1j, position students to identify and use coordinate plane vocabulary accurately by asking questions like:

- Which point is the x-coordinate? How do you know?
- Which point is the y-coordinate? How do you know?
- What do you notice the x-values increasing by?
- What do you notice the y-values increasing by?
- What does that mean from the context of the problem?

1.1.3 GUIDED PRACTICE: INVESTIGATING PATTERNS

Component Overview: Students analyze patterns in cost to create equations and a graphical representation of the pattern.

DIFFERENTIATED INSTRUCTION

As a visual entry point, consider encouraging students who are having difficulty getting started or noticing the pattern to draw a visual model of how much each container costs. This practice may also help students recognize that every two containers costs one dollar.

TEACHER MOVES

Notice how students can think of the relationship between containers and cost in two ways: two containers cost one dollar, or every container costs half a dollar. Consider engaging in a discussion on why both relationships are true and how using the relationship of every container costing half a dollar is helpful when writing the equation in question 1b.

ENRICHMENT

Position students to connect the cost pattern to the points on the coordinate plane by asking:

- How can you see the relationship of containers to cost on the coordinate plane?
- How did you decide which was the x-value and which was the y-value?



1.1.3 Guided Practice: Investigating Patterns

- The orange juice manufacturer decided to offer orange juice in pint-sized containers. Each 8-pack of pint-sized containers will be sold for \$4.

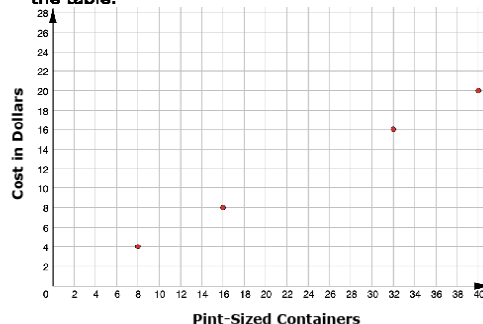
- Use the pattern in the table to find the missing values.

Pint-Sized Containers	Cost (\$)
8	4
16	8
32	16
40	20

- Write an equation to represent the relationship between the number of pint-sized containers, x , and the cost in dollars, y .

$$y = \frac{1}{2}x$$

- Use the coordinate grid to plot the ordered pairs in the table.



- d. Use the table, the equation, or the graph to determine the cost of 24 pint-sized containers.
\$12

- e. Which representation did you use?
Answers will vary.

2. Elena, Kimiko, and Vicente were discussing the two graphs.

Vicente: In the first graph, would the ordered pair (2.5, 25) fit the pattern?

Elena: I don't think so. Having 2.5 cartons does not make sense.

Kimiko: The ordered pair (15, 7.5) for the second graph does make sense. That would be 15 pint-size containers costing \$7.50.

Elena: No, Kimiko, the 15 is \$15, and the 7.5 is $7\frac{1}{2}$ pint-size containers. A $\frac{1}{2}$ pint-size container would not make sense.

- a. Explain whether or not you think the ordered pairs (2.5, 25) and (15, 7.5) make sense for the contexts they represent.
Answers will vary. (2.5, 25) does make not sense because the x-coordinate, 2.5, represents 2.5 cartons which is not possible. (15, 7.5) does make sense because that ordered pair represents 15 cartons costing \$7.50. Some students may argue that (15, 7.5) does NOT make sense because the juices are sold in packs of 8 and 15 is not a multiple of 8.
- b. Ask your partner to explain their answer to you. Write down their explanation.
Answers will vary.

1.1.3 GUIDED PRACTICE: INVESTIGATING PATTERNS (CONTINUED)

TEACHER MOVES

To support mathematical discussion and interaction with the reasoning of others, consider having students act as Elena, Kimiko, and Vicente to help understand and compare the reasoning of each. Then as a class, have students share the answers to 2a and 2b as each of these students. This discussion-based and auditory approach can also be used in a teacher-led small group for students who may need additional scaffolding.

ENRICHMENT

For question 2a, position students to identify the relationship for (15, 7.5) as 7.5 being half of 15 and how this reasoning can be used to explain why the coordinate pair (2.5, 25) does not make sense. Ask questions like:

- What is the relationship between 15 containers and the cost of \$7.50?
- How is this similar to the relationships in the table on 1a? Is this relationship true for (2.5, 25)? Why or why not?

DIFFERENTIATED INSTRUCTION

If students have trouble writing what their partner says, encourage them to paraphrase their partner's words, then write, and then have their partner read their written response to see if it has the same content.

1.1.4 WRAP-UP: REFLECTION

Component Overview: Students self-assess their confidence to use expressions to extend patterns, locate ordered pairs on a coordinate plane, identify the pattern, and write expressions to match. Have students complete this self-assessment individually as a reflection on learning from the lesson.

DIFFERENTIATED INSTRUCTION

If students are having difficulty assessing their learning, prompt students to revisit the parts of the lesson on which each statement was learned or practiced. The statement pairs correlate to:

- Pair 1: 1.1.2 Guided Instruction
- Pair 2: 1.1.3 Guided Practice
- Pair 3: 1.1.3 Guided Practice



1.1.4 Wrap-Up: Reflection

1. For each pair of statements, check the statements that apply to your understanding.

Answers will vary.

- ☐ I can use an expression to extend patterns.
- ☐ I need support using an expression to extend patterns.

- ☐ I can locate ordered pairs on a coordinate plane.
- ☐ I need support in locating ordered pairs on a coordinate plane.

- ☐ I can identify a pattern and write an expression to match.
- ☐ I need support identifying a pattern and writing an expression to match.

